Cell Biology and Genetics

1. Introduction

Cell biology is the study of the structure, function, and behavior of cells, which are the basic structural and functional units of life. Genetics is the study of heredity, variation, genes, and the transmission of traits from one generation to the next.

In zoology, these subjects are especially important because they explain how animal cells work, how animal traits are inherited, how genetic information is stored and expressed, and how variation arises in populations.

2. Importance of the topic

Cell biology and genetics form the foundation of modern zoology, biotechnology, medicine, and evolutionary biology. They help students understand disease mechanisms, cell division, reproduction, mutations, inheritance, and molecular regulation.

These subjects are also necessary for practical zoology because many experiments involve microscopic study of cell division, chromosome behavior, and genetic inheritance patterns.

3. Cell theory

Cell theory is the basic concept of cell biology. It states that all living organisms are made of cells, the cell is the basic unit of structure and function, and new cells arise from pre-existing cells.

4. Types of cells

Cells are broadly classified into prokaryotic and eukaryotic cells. Animal cells are eukaryotic and have a true nucleus and membrane-bound organelles.

Prokaryotic cells

- No true nucleus.
- No membrane-bound organelles.
- DNA is located in nucleoid region.

Eukaryotic cells

- True nucleus present.
- Membrane-bound organelles present.
- Larger and more complex than prokaryotic cells.

5. Structure of animal cell

An animal cell consists of plasma membrane, cytoplasm, nucleus, and various organelles. Each organelle has a specific function in maintaining cellular life.

Main parts

- Plasma membrane.
- Cytoplasm.
- Nucleus.
- Mitochondria.
- Endoplasmic reticulum.
- Golgi complex.
- Lysosomes.
- Ribosomes.
- Centrioles.
- Cytoskeleton.

6. Plasma membrane

The plasma membrane forms the outer boundary of the cell and controls the entry and exit of substances. It is selectively permeable and helps maintain internal balance.

Functions

- Protection.
- Selective transport.
- Cell signalling.
- Cell recognition.
- Endocytosis and exocytosis.

7. Endoplasmic reticulum

The endoplasmic reticulum is a membrane network involved in synthesis and transport. Rough ER has ribosomes and helps in protein synthesis, while smooth ER helps in lipid synthesis and detoxification.

8. Golgi complex

The Golgi complex modifies, packages, and secretes proteins and lipids. It also helps form secretory vesicles and lysosomes.

9. Lysosomes

Lysosomes contain digestive enzymes and are involved in intracellular digestion, autophagy, and removal of worn-out cell parts. They are often called the “suicidal bags” of the cell.

10. Mitochondria

Mitochondria are the sites of aerobic respiration and ATP production. Because they generate energy for the cell, they are often called the powerhouses of the cell.

11. Ribosomes

Ribosomes are the sites of protein synthesis. They may be free in the cytoplasm or attached to rough ER.

12. Nucleus

The nucleus controls the activities of the cell and stores genetic material in the form of DNA. It contains chromatin, nucleoplasm, nucleolus, and nuclear membrane.

Functions

- Genetic control.
- DNA storage.
- RNA synthesis.
- Regulation of cell division.

13. Chromosomes

Chromosomes are thread-like structures made of DNA and proteins. They carry genes and become clearly visible during cell division.

Important terms

- Chromatin.
- Chromosome.
- Chromatid.
- Centromere.
- Telomere.
- Gene locus.

14. Cell division

Cell division is the process by which one cell produces new cells. It is essential for growth, repair, reproduction, and continuity of life.

Types

- Mitosis.
- Meiosis.
- Amitosis in some simple forms.

15. Mitosis

Mitosis is an equational division in which one parent cell produces two daughter cells genetically identical to the parent cell. It occurs in somatic cells.

Phases of mitosis

1. Prophase.
2. Metaphase.
3. Anaphase.
4. Telophase.
5. Cytokinesis.

Importance

- Growth.
- Repair.
- Asexual reproduction in some organisms.
- Maintenance of chromosome number.

16. Meiosis

Meiosis is a reductional division in which one diploid cell produces four haploid cells. It occurs in germ cells and is essential for sexual reproduction.

Stages

- Meiosis I: reduction division.
- Meiosis II: equational division.

Importance

- Formation of gametes.
- Genetic variation.
- Maintenance of chromosome number across generations.

17. Gametogenesis

Gametogenesis is the formation of gametes. In animals, spermatogenesis produces sperm and oogenesis produces ova.

18. Parthenogenesis

Parthenogenesis is the development of an organism from an unfertilized egg. It is a special reproductive strategy in some animals.

19. Molecular biology

Molecular biology explains the structure and function of nucleic acids and how genetic information is stored, copied, and expressed.

20. DNA structure

DNA, or deoxyribonucleic acid, is the hereditary material in most animals. It consists of two antiparallel polynucleotide strands forming a double helix.

Components of DNA

- Sugar: deoxyribose.
- Phosphate group.
- Nitrogenous bases: adenine, thymine, guanine, cytosine.

Base pairing

- A Pairs with T.
- G Pairs with C.

21. DNA replication

DNA replication is the process by which DNA makes an identical copy of itself before cell division. It is semi-conservative.

Main features

- Starts at origin.
- Uses template strands.
- Involves enzymes such as helicase and DNA polymerase.
- Produces two identical DNA molecules.

22. RNA structure and types

RNA, or ribonucleic acid, is usually single-stranded and plays a key role in protein synthesis.

Types of RNA

- mRNA: messenger RNA.
- tRNA: transfer RNA.
- rRNA: ribosomal RNA.

Importance

RNA transfers genetic information from DNA and helps build proteins.

23. Protein synthesis

Protein synthesis is the process by which cells make proteins using genetic information. It has two main stages: transcription and translation.

Transcription

DNA is used as a template to make mRNA.

Translation

mRNA is read by ribosomes to assemble amino acids into a polypeptide chain.

24. Genetic code

The genetic code is the set of rules by which nucleotide triplets in mRNA specify amino acids. It is read in codons.

Properties of genetic code

- Triplet code.
- Universal.
- Degenerate.
- Non-overlapping.
- Commaless.
- Read in a fixed direction.

25. Gene regulation and operon concept

Gene expression is controlled so that cells produce the right proteins at the right time. In prokaryotes, the operon model explains how groups of genes are regulated together.

Operon parts

- Promoter.
- Operator.
- Structural genes.
- Regulator gene.
- Repressor.
- Inducer.

Importance

Gene regulation ensures efficient use of energy and proper cellular function.

26. Mendelian genetics

Mendelian genetics explains inheritance based on the laws discovered by Gregor Mendel.

Mendel’s laws

- Law of dominance.
- Law of segregation.
- Law of independent assortment.

Key terms

- Gene.
- Allele.
- Genotype.
- Phenotype.
- Homozygous.
- Heterozygous.

27. Non-Mendelian inheritance

Non-Mendelian inheritance includes patterns that do not follow simple Mendelian ratios. These may involve incomplete dominance, codominance, multiple alleles, polygenic inheritance, cytoplasmic inheritance, and gene linkage.

28. Linkage and crossing over

Linked genes are located on the same chromosome and tend to be inherited together. Crossing over occurs during meiosis and exchanges genetic material between homologous chromosomes, creating variation.

29. Sex determination and sex-linked inheritance

Sex determination is the mechanism by which the sex of an organism is established. Sex-linked inheritance refers to traits controlled by genes on sex chromosomes.

Examples

- X-linked traits.
- Y-linked traits.
- Sex-linked color blindness and hemophilia in classical genetics examples.

30. Mutation

Mutation is a sudden heritable change in genetic material. It may occur in genes or chromosomes and can arise spontaneously or be induced by mutagens.

Types of chromosomal mutations

- Deletion.
- Duplication.
- Inversion.
- Translocation.
- Aneuploidy.
- Polyploidy.

Gene mutation

A change in the DNA sequence of a gene.

31. Inborn errors of metabolism

Inborn errors of metabolism are hereditary disorders caused by defective enzymes or metabolic pathways. They show the link between genes and biochemical function.

32. One gene-one enzyme and one gene-one polypeptide

These concepts explain how genes control the synthesis of proteins and enzymes. A gene can determine the production of a specific enzyme or polypeptide chain.

33. Applications in zoology

Cell biology and genetics are useful in:

- Animal breeding.
- Disease diagnosis.
- Biotechnology.
- Conservation genetics.
- Wildlife management.
- Reproductive biology.
- Research in developmental biology.
- Evolutionary studies.

34. Summary

Cell biology and genetics are the foundation of zoology because they explain how animal cells are built, how they divide, how genetic information is stored, and how traits are inherited. Together, these subjects connect structure, function, heredity, and molecular control in living animals.